

A 0.45 Upper Bound for the i.i.d. Berry–Esseen Constant

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Abstract

We prove that the absolute constant in the classical Berry–Esseen inequality for identically distributed summands is at most 0.45. The key quantity is the symmetrized third-moment ratio $r = \mathbb{E}|X - X'|^3 / (2\mathbb{E}|X|^3)$, where X' is an independent copy of a centered variance-one summand X . Symmetrization yields a modulus bound at scale $r\mathbb{E}|X|^3$, while the same r sharpens the third-order bound for the imaginary part of the characteristic function. Retaining this dependence gives a sharper single-summand comparison with the Gaussian characteristic function. Prawitz smoothing then reduces the proof to a scalar inequality on a compact parameter domain, which is established by exact dyadic interval certificates. The analytic reduction and the exact certificate are formally verified in Lean.

1 Introduction

Let $X, X_1, X_2, \dots$ be independent and identically distributed real random variables with

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \rho := \mathbb{E}|X|^3 < \infty,$$

and call a law satisfying these conditions admissible. Set

$$S_n := \frac{X_1 + \dots + X_n}{\sqrt{n}}, \quad \Delta_n := \sup_{x \in \mathbb{R}} |\mathbb{P}\{S_n \leq x\} - \Phi(x)|,$$

where Φ denotes the standard normal distribution function. The optimal i.i.d. Berry–Esseen constant is

$$C_0 := \inf \left\{ C > 0 : \Delta_n \leq C \frac{\rho}{\sqrt{n}} \text{ for all admissible laws and all } n \geq 1 \right\}.$$

Esseen’s classical example gives

$$C_0 \geq C_E := \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = 0.409732\dots [3].$$

Shevtsova proved the published upper bound $C_0 \leq 0.4690$ [7]; Bobkov’s survey records the resulting interval $0.4097 < C_0 < 0.4690$ [1, Theorem 20.1]. To the best of our knowledge, 0.4690 is the smallest previously published general i.i.d. upper bound.

Theorem 1. *For every admissible law and every integer $n \geq 1$,*

$$\Delta_n \leq 0.45 \frac{\rho}{\sqrt{n}}.$$

Consequently, $C_0 \leq 0.45$.

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The improvement comes from retaining the dependence between two estimates for the characteristic function. If X' is an independent copy of X , set

$$r := \frac{\mathbb{E}|X - X'|^3}{2\rho}.$$

The symmetrized law gives a modulus bound depending on $r\rho$. A stop-loss identity, applied to the same ratio, bounds the sine remainder $\mathbb{E}(\sin(uX) - uX)$, equivalently the imaginary part of $\mathbb{E}e^{iuX} - (1 - u^2/2)$. The modulus bound weakens as r increases, whereas the bound on the sine remainder strengthens. Preserving this dependence sharpens the comparison between $\mathbb{E}e^{iuX}$ and $e^{-u^2/2}$.

Write $L = \rho/\sqrt{n}$. A direct estimate handles $L \geq 56/45$. For $L < 56/45$, Prawitz's smoothing inequality turns these characteristic-function bounds into an explicit scalar inequality in (n, ρ, r) . Exact interval arithmetic proves it on the full continuous parameter domain.

2 Moment constraints for the characteristic function

Two pointwise estimates will be needed below: a modulus bound obtained from $X - X'$, and a bound on the sine remainder $\mathbb{E}(\sin(uX) - uX)$. Both depend on the ratio r . We first determine its admissible range and the corresponding restriction on the sine remainder.

2.1 The symmetrized third-moment ratio

Let X' be an independent copy of X , and define

$$m := \mathbb{E}|X - X'|^3, \quad r := \frac{m}{2\rho}.$$

For $t \geq 0$, introduce the stop-loss functions

$$A(t) := \mathbb{E}(X - t)_+, \quad B(t) := \mathbb{E}(-X - t)_+,$$

and put $r_0 := \mathbb{E}X_+ = \mathbb{E}X_-$.

Lemma 2. *One has*

$$m = 2\rho + 12 \int_0^\infty (A(t)^2 + B(t)^2) dt, \tag{2.1}$$

$$3 \int_0^\infty A(t)^2 dt \leq r_0 \mathbb{E}X_+^2, \quad 3 \int_0^\infty B(t)^2 dt \leq r_0 \mathbb{E}X_-^2. \tag{2.2}$$

Consequently,

$$1 \leq r \leq 1 + \frac{\mathbb{E}|X|}{\rho} \leq 1 + \frac{1}{\rho} \leq 2. \tag{2.3}$$

If $\mu := \mathbb{E}X^3$, then

$$(r - 1)^2 + \left(\frac{\mu}{\rho}\right)^2 \leq 1. \tag{2.4}$$

Proof. For real x, y ,

$$|x - y|^3 = 6 \int_{\mathbb{R}} [(x - t)_+(t - y)_+ + (y - t)_+(t - x)_+] dt.$$

Taking expectations, using independence, splitting the integral at zero, and applying $\mathbb{E}X = 0$ gives

$$m = 12 \int_0^\infty (A^2 + B^2 + tA + tB) dt.$$

Fubini's theorem gives $\int_0^\infty tA(t) dt = \mathbb{E}X_+^3/6$ and the analogous identity for B , proving (2.1).

If $0 \leq y \leq z$, direct integration gives

$$3 \int_0^\infty (y-t)_+(z-t)_+ dt = \frac{3}{2}y^2z - \frac{1}{2}y^3 \leq \frac{1}{2}(yz^2 + y^2z).$$

Averaging independent copies proves (2.2). Since $\mathbb{E}X_+^2 + \mathbb{E}X_-^2 = 1$, it follows that $m - 2\rho \leq 4r_0 = 2\mathbb{E}|X|$, and (2.3) follows from $\mathbb{E}|X| \leq 1$ and Lyapunov's inequality $\rho \geq 1$.

To prove (2.4), write

$$a_j := \mathbb{E}X_+^j, \quad b_j := \mathbb{E}X_-^j \quad (j = 2, 3).$$

Positive semidefiniteness of the covariance matrix of (X_+, X_-) , together with $X_+X_- = 0$, gives $a_2b_2 \geq r_0^2$. Cauchy's inequality gives $a_2^2 \leq r_0a_3$ and $b_2^2 \leq r_0b_3$, hence $r_0 \leq \sqrt{a_3b_3}$. Therefore

$$m - 2\rho \leq 4\sqrt{a_3b_3} = 2\sqrt{\rho^2 - \mu^2}.$$

Division by 2ρ proves (2.4). $\square$

The range in (2.3) sets the scale in the modulus estimate. At the same time, (2.4) shows that a larger value of $r - 1$ forces the normalized third moment to be smaller. The next result transfers this restriction to the sine remainder at every frequency.

2.2 A bound for the sine remainder

Since $\mathbb{E}X = 0$, the imaginary part of $\mathbb{E}e^{iuX} - (1 - u^2/2)$ is $\mathbb{E}(\sin(uX) - uX)$. The same ratio r controls this quantity uniformly in u .

Lemma 3 (Bound for the sine remainder). *For every $u \in \mathbb{R}$,*

$$|\mathbb{E}(\sin(uX) - uX)| \leq \frac{\rho|u|^3}{6} \sqrt{1 - (r - 1)^2}. \quad (2.5)$$

Proof. It is enough to take $u > 0$. Define

$$J := \int_0^\infty t(A + B) dt = \frac{\rho}{6}, \quad E_0 := \int_0^\infty (A^2 + B^2) dt, \quad H_u := \frac{1}{u} \int_0^\infty \sin(ut)(A - B) dt.$$

For a centered two-point law, these normalized quantities satisfy $E_0^2 + H_u^2 \leq J^2$. Stop-loss functions are affine under mixtures, so Minkowski's inequality carries the same constraint to an arbitrary centered law. We now make the mixture explicit.

Let α_+ be the restriction of the law of X to $(0, \infty)$, and let α_- be the reflection of its restriction to $(-\infty, 0)$. Centering gives

$$\int a \alpha_+(da) = \int b \alpha_-(db) = r_0,$$

and the variance-one assumption implies $r_0 > 0$. On $(0, \infty)^2$, define

$$\nu(da, db) := \frac{a+b}{r_0} \alpha_+(da) \alpha_-(db),$$

and add mass $\mathbb{P}\{X = 0\}$ at a zero component. For each (a, b) , let $X_{a,b}$ take the values a and $-b$ with probabilities $b/(a+b)$ and $a/(a+b)$. A direct calculation shows that the law of X is the ν -mixture of these centered two-point laws and the zero law. In particular, their stop-loss functions mix linearly to (A, B) .

For one such component, let $J_{a,b}, E_{a,b}, H_{a,b}$ denote the three quantities above, and put $x = ua, y = ub$. Direct calculation gives

$$\frac{E_{a,b}}{J_{a,b}} = \frac{2xy}{x^2 + y^2}, \quad \frac{H_{a,b}}{J_{a,b}} = -\frac{6(y \sin x - x \sin y)}{xy(x^2 + y^2)}.$$

For $z > 0$,

$$\left| \frac{d}{d(z^2)} \frac{\sin z}{z} \right| = \frac{|z \cos z - \sin z|}{2z^3} \leq \frac{1}{6},$$

because

$$|z \cos z - \sin z| = \left| \int_0^z t \sin t \, dt \right| \leq \int_0^z t^2 \, dt = \frac{z^3}{3}.$$

It follows that $E_{a,b}^2 + H_{a,b}^2 \leq J_{a,b}^2$. Jensen's inequality applied at each t , together with linearity of the mixture, gives

$$E_0 \leq \int E_{a,b} \nu(da, db), \quad H_u = \int H_{a,b} \nu(da, db), \quad J = \int J_{a,b} \nu(da, db).$$

Minkowski's inequality in $\mathbb{R}^2$ now proves

$$\sqrt{E_0^2 + H_u^2} \leq \int \sqrt{E_{a,b}^2 + H_{a,b}^2} \nu(da, db) \leq J.$$

Two integrations of the sine remainder yield $\mathbb{E}(uX - \sin(uX)) = u^3 H_u$. By (2.1), $E_0/J = r - 1$. Thus $E_0^2 + H_u^2 \leq J^2$ gives $|H_u| \leq (\rho/6)\sqrt{1 - (r - 1)^2}$, proving (2.5). $\square$

Thus the symmetrized third-moment ratio governs both quantities needed below: it sets the scale of the modulus estimate and sharpens the bound on the sine remainder.

3 Comparison with the Gaussian characteristic function

Let $f(u) := \mathbb{E}e^{iuX}$. At each frequency, the smoothing argument will use both a bound for $|f(u)|$ and a bound for $|f(u) - e^{-u^2/2}|$. We derive both in terms of the same ratio r .

For the modulus bound, put

$$g(v) := \cos v - 1 + \frac{v^2}{2}, \quad \kappa := \max_{v>0} \frac{g(v)}{v^3}.$$

Exact alternating-series bounds and interval Newton verification give

$$0.09916191350 < \kappa < 0.09916191353$$

and a unique maximizer ϑ satisfying

$$3.99589567 < \vartheta < 3.99589570.$$

We use Prawitz's piecewise characteristic-function minorant [6], retaining the scale $r\rho$ controlled in Section 2:

$$q(v) := \begin{cases} \frac{1}{2} - \kappa v, & 0 \leq v \leq \vartheta, \\ \frac{1 - \cos v}{v^2}, & \vartheta < v \leq 2\pi, \\ 0, & v > 2\pi. \end{cases} \quad (3.1)$$

Lemma 4. *The function q is convex and nonincreasing, and*

$$q(v) \leq \frac{1 - \cos v}{v^2} \quad (v > 0).$$

Consequently,

$$|f(u)|^2 \leq 1 - 2u^2 q(r\rho|u|). \quad (3.2)$$

Proof. The definition of κ gives the minorant on the first interval. At ϑ , equality and equality of the first derivatives hold. For $\psi(v) = (1 - \cos v)/v^2$,

$$\psi''(v) = \frac{v^2 \cos v - 4v \sin v + 6(1 - \cos v)}{v^4}.$$

The numerator is nonnegative on $[\vartheta, 2\pi]$, and $\psi'(2\pi) = 0$. The required breakpoint and sign statements are verified exactly in the Lean development. Hence the three pieces join with nondecreasing slopes and q is convex and nonincreasing.

Let $Y = X - X'$. Since $\mathbb{E}Y^2 = 2$, the measure $Y^2 d\mathbb{P}/2$ is a probability measure. Jensen's inequality gives

$$\begin{aligned} 1 - |f(u)|^2 &= \mathbb{E}(1 - \cos(uY)) \\ &\geq 2u^2 q \left(|u| \frac{\mathbb{E}|Y|^3}{\mathbb{E}Y^2} \right) = 2u^2 q(r\rho|u|), \end{aligned}$$

which is (3.2). □

We next bound the difference from the Gaussian characteristic function. After normalization by $\rho|u|^3$, the ordinary third-order Taylor bound places the remainder in a disk of radius $1/6$, while Lemma 3 restricts its imaginary part according to r . The corresponding normalized Gaussian correction lies on the real axis, so the largest possible distance is a two-dimensional constrained maximum. Set

$$\beta(r) := \frac{1}{6} \sqrt{1 - (r-1)^2}, \quad d_0 := \frac{1}{6},$$

and, for $c \geq 0$, define

$$\epsilon_\rho(c) := \begin{cases} \frac{\rho^2}{c^3} \left(e^{-c^2/(2\rho^2)} - 1 + \frac{c^2}{2\rho^2} \right), & c > 0, \\ 0, & c = 0. \end{cases}$$

Let

$$\Xi(r) := \{0, \kappa\} \cup \left(\left\{ \frac{r-1}{6} \right\} \cap [0, \kappa] \right)$$

and

$$D(\rho, r, c)^2 := \max_{x \in \Xi(r)} \left\{ (x - \epsilon_\rho(c))^2 + \min(\beta(r)^2, d_0^2 - x^2) \right\}. \quad (3.3)$$

Thus $D(\rho, r, c)$ is the greatest possible distance from the normalized Gaussian correction $\epsilon_\rho(c)$ to a Taylor remainder satisfying the disk and imaginary-part bounds; $\Xi(r)$ lists the only real parts at which the maximum can occur.

Lemma 5. *For every $u \in \mathbb{R}$, with $c = \rho|u|$,*

$$|f(u) - e^{-u^2/2}| \leq \rho|u|^3 D(\rho, r, c). \quad (3.4)$$

Proof. The case $u = 0$ is immediate. Suppose $u \neq 0$, and set

$$z := \frac{f(u) - (1 - u^2/2)}{\rho|u|^3},$$

the definition of κ , Lemma 3, and the third-order Taylor remainder give

$$0 \leq \Re z \leq \kappa, \quad |\Im z| \leq \beta(r), \quad |z| \leq d_0.$$

Moreover,

$$\frac{e^{-u^2/2} - (1 - u^2/2)}{\rho|u|^3} = \epsilon_\rho(c).$$

Thus the normalized left side of (3.4) is the distance from the nonnegative real point $\epsilon_\rho(c)$ to the intersection of a vertical strip, a horizontal strip, and the disk of radius d_0 . For fixed real part x , the largest squared imaginary part is $\min(\beta(r)^2, d_0^2 - x^2)$. The squared distance can change form only at $x = (r - 1)/6$, and its maximum on each piece is attained at an endpoint. These are exactly the points in $\Xi(r)$, proving (3.3) and (3.4). $\square$

The two estimates needed for smoothing are now explicit: (3.2) bounds the modulus, and (3.4) bounds the deviation from the Gaussian, with their dependence on the same r preserved.

4 Prawitz smoothing and the 0.45 bound

Set $L := \rho/\sqrt{n}$. When L is large, a distribution-free Kolmogorov bound is sufficient. When L is smaller, Prawitz smoothing turns the two characteristic-function estimates from Section 3 into the desired distributional bound. We begin with the large- L regime.

Lemma 6. *If W is centered and has variance one, then*

$$\sup_x |\mathbb{P}\{W \leq x\} - \Phi(x)| \leq \frac{14}{25} = 0.56.$$

Proof. Cantelli's inequality gives, for $a \geq 0$,

$$\mathbb{P}\{W \geq a\} \leq \frac{1}{1 + a^2}, \quad \mathbb{P}\{W \leq -a\} \leq \frac{1}{1 + a^2}.$$

Hence the Kolmogorov distance is at most

$$\max \left\{ \frac{1}{2}, \sup_{a \geq 0} \left(\frac{1}{1 + a^2} - \Phi(-a) \right) \right\}.$$

For $a \geq 1$, the second term is at most $1/2$. For $0 \leq a \leq 1$, the bound $(2\pi)^{-1/2} < 2/5$ yields

$$\frac{1}{1 + a^2} - \Phi(-a) \leq \frac{1}{1 + a^2} - \frac{1}{2} + \frac{2a}{5}.$$

The last expression is at most $14/25$ exactly when

$$3 - 20a + 53a^2 - 20a^3 \geq 0.$$

On $[0, 1/4]$ the left side is at least $3 - 20a + 48a^2 > 0$, and on $[1/4, 1/2]$ it is at least $3 - 20a + 43a^2 > 0$. On $[1/2, 1]$, write the polynomial as $3 + a(-20a^2 + 53a - 20)$; the concave quadratic in parentheses is positive at both endpoints. This proves the claim. $\square$

Applied to $W = S_n$, Lemma 6 proves Theorem 1 whenever $L \geq 56/45$, since $0.56 = 0.45(56/45)$. It remains to consider

$$1 \leq \rho < \frac{56}{45} \sqrt{n}.$$

For this range we use the following form of the Prawitz smoothing inequality. For $0 < t < 1$, define the Prawitz kernel

$$\begin{aligned} K(t) &:= \frac{1-t}{2} + \frac{i}{2} \left((1-t) \cot(\pi t) + \frac{1}{\pi} \right), \\ K(-t) &:= \overline{K(t)}. \end{aligned} \tag{4.1}$$

The following normalization of the Prawitz smoothing inequality converts characteristic-function estimates into a bound on Kolmogorov distance [5]. It is equivalent to the filter normalization recorded in [4, Section 2].

Lemma 7 (Prawitz smoothing inequality). *Let Y be an integrable real random variable, with distribution function F and characteristic function h . For $T > 0$ and $0 < t_0 < 1$,*

$$\begin{aligned} \sup_x |F(x) - \Phi(x)| &\leq 2 \int_0^{t_0} |K(t)| |h(Tt) - e^{-T^2 t^2/2}| dt \\ &\quad + 2 \int_{t_0}^1 |K(t)| |h(Tt)| dt \\ &\quad + 2 \int_0^{t_0} \left| K(t) - \frac{i}{2\pi t} \right| e^{-T^2 t^2/2} dt \\ &\quad + \frac{1}{\pi} \int_{t_0}^\infty e^{-T^2 t^2/2} \frac{dt}{t}. \end{aligned} \tag{4.2}$$

Proof. Let $J(x) = (\sin(\pi x)/(\pi x))^2$, with its removable value at zero, and let H be the odd band-limited function in Prawitz's construction. The pointwise estimate $|\operatorname{sgn}(x) - H(x)| \leq J(x)$ gives

$$B_-(x) := \frac{1 - H(x) - J(x)}{2} \leq \mathbf{1}_{(-\infty, 0]}(x) \leq \frac{1 - H(x) + J(x)}{2} =: B_+(x).$$

The nonconstant Fourier transforms of B_+ and B_- are supported on $[-1, 1]$; under the normalization in (4.1), they are $K(t)$ and $-K(-t)$, respectively.

Apply these bounds to $T(Y - x)/(2\pi)$, take expectations, and use Fourier inversion. The finite first moment removes the apparent singularity at zero:

$$|h(u) - 1| \leq |u| \mathbb{E}|Y|,$$

so the zero-frequency constant may be subtracted before passing to the symmetric Fourier limit. Subtracting the exact normal inversion formula, splitting the frequencies at t_0 , and using conjugate symmetry gives the first three terms in (4.2); the remaining normal frequencies give the last integral. Applying the same argument to $-Y$ supplies the opposite CDF inequality and completes the absolute-value bound. $\square$

We apply Lemma 7 to the law of S_n , which is integrable under the third-moment assumption. With $L = \rho/\sqrt{n}$ as above, choose

$$T := \frac{2\pi}{rL}, \quad t_0 := \frac{19}{50}. \tag{4.3}$$

For $0 \leq t \leq 1$, set

$$\begin{aligned} u(t) &:= \frac{2\pi t}{r\rho}, \\ a(t) &:= [1 - 2u(t)^2 q(2\pi t)]_+^{1/2}, & b(t) &:= e^{-u(t)^2/2}, \\ M(t) &:= a(t)^n, & N(t) &:= b(t)^n = e^{-T^2 t^2/2}, \end{aligned}$$

and

$$R(t) := \min \left\{ n\rho u(t)^3 D \left(\rho, r, \frac{2\pi t}{r} \right) \max(a(t), b(t))^{n-1}, M(t) + N(t) \right\}.$$

Here $a(t)$ and $b(t)$ are the single-summand characteristic-function and Gaussian bounds; $M(t)$ and $N(t)$ are their n -fold counterparts. The two entries in the minimum defining

$R(t)$ are, respectively, the bound obtained from the difference-of-powers identity and the triangle-inequality bound. With these bounds, the right-hand side of (4.2) becomes

$$\begin{aligned} \mathcal{U}(n, \rho, r) := & 2 \int_0^{t_0} |K(t)| R(t) dt + 2 \int_{t_0}^1 |K(t)| M(t) dt \\ & + 2 \int_0^{t_0} \left| K(t) - \frac{i}{2\pi t} \right| N(t) dt + \frac{1}{\pi} \int_{t_0}^\infty N(t) \frac{dt}{t}. \end{aligned} \quad (4.4)$$

Lemma 8 (Exact scalar bound). *For every integer $n \geq 1$, every*

$$1 \leq \rho \leq \frac{56}{45} \sqrt{n}, \quad 1 \leq r \leq 1 + \frac{1}{\rho}, \quad (4.5)$$

one has

$$\mathcal{U}(n, \rho, r) \leq 0.4495 \frac{\rho}{\sqrt{n}}. \quad (4.6)$$

This is the only exhaustive computer-assisted step. Section 5 proves it by exact interval arithmetic over the entire domain (4.5).

Proof of Theorem 1. If $L \geq 56/45$, Lemma 6 gives $\Delta_n \leq 0.56 \leq 0.45L$. Suppose $L < 56/45$. Then Lemma 2 places (n, ρ, r) in (4.5).

The characteristic function of S_n is $f_n(t) = f(t/\sqrt{n})^n$. With the choices in (4.3), Lemma 4 gives

$$|f(u(t))| \leq a(t), \quad |f_n(Tt)| \leq M(t).$$

Lemma 5 and $|z^n - w^n| \leq n|z - w| \max(|z|, |w|)^{n-1}$ give the first bound in the definition of $R(t)$; the triangle inequality gives the second. Lemma 7 therefore yields

$$\Delta_n \leq \mathcal{U}(n, \rho, r).$$

Lemma 8 now gives

$$\Delta_n \leq 0.4495L < 0.45L,$$

which completes the proof. $\square$

5 Exact verification of the scalar inequality

Lemma 8 is proved by a finite exact certificate. The certificate represents a cover of the continuous parameter domain, and a proved-sound interval checker establishes the required upper bound on every cell of that cover.

5.1 Exact interval bounds

All numerical endpoints are dyadic rationals at precision 2^{-48} . Addition, subtraction, multiplication, division, squaring, nonnegative square root, and integer powers use proved outward-rounding rules. Alternating Taylor polynomials of degrees 24 and 25, together with exact range reduction, enclose the exponential and trigonometric functions used by the checker. The exponential-integral tail is bounded by a convex composite trapezoid rule on $[0, 24]$ and a proved infinite-tail estimate.

Every integral in (4.4) is replaced by an upper Darboux sum. Lean proves the generic implication

$$\text{cellwise interval containment} \implies \int_J F(t) dt \leq \sum_{I \in \mathcal{P}} |I| \sup[F](I),$$

including the outward-rounded accumulation used by the interval checker. Removable singularities at $t = 0$ and $t = 1$ are handled algebraically before interval evaluation.

5.2 Covering the continuous domain

For each $1 \leq n < 100$, the parameter domain is mapped to the unit square by

$$\rho = 1 + y \left(\frac{56}{45} \sqrt{n} - 1 \right), \quad r = 1 + \frac{z}{\rho}, \quad 0 \leq y, z \leq 1.$$

A prefix code describes a binary subdivision tree. A leaf is accepted only when the exact interval upper bound is below 0.4495. Lean proves that an accepted tree covers the entire square and that every represented point satisfies (4.6).

For $n \geq 100$, write $L = \rho/\sqrt{n}$. The bounds $a(t), b(t) \leq e^{-u(t)^2 q(2\pi t)}$, together with $(n-1)/n \geq 99/100$, give

$$\max(a(t), b(t))^{n-1} \leq \exp \left\{ -\frac{99}{100} \frac{(2\pi t)^2 q(2\pi t)}{r^2 L^2} \right\}.$$

This removes the explicit n -dependence from the telescoping term. Monotonicity and convexity reduce the remaining dependence on ρ . The checker covers three compact regions: $0 \leq L \leq 1/16$, $1 \leq r \leq 2$; $1/16 \leq L \leq 1/10$, $1 \leq r \leq 2$; and $1/10 \leq L \leq 56/45$, $1 \leq r \leq 1 + (10L)^{-1}$. Since $\rho \geq 1$, $n \geq 100$, and $r \leq 1 + 1/\rho$, their union contains every admissible pair. The endpoint $L = 0$ uses a proved removable-limit formula. Checker soundness then gives (4.6) throughout each accepted covering tree.

In the upper region, the interval parameter map is intersected with the proved constraint $r \leq 2$ before its tree is applied, so the endpoint is covered without enlarging the admissible domain.

The complete Lean source and the exact certificate data are available at github.com/haonan-xiao/iid-berry-esseen.

The probability-theoretic reduction, moment identities, characteristic-function estimates, smoothing argument, checker soundness, and covering theorems are formalized in Lean 4 [2] on top of Mathlib [8]. Only the prefix codes, not the program that searches for them, enter the proof. Lean recomputes every accepted closed Boolean certificate with `native_decide`. The preceding formal arguments are kernel-checked; certificate evaluation additionally trusts Lean’s native compiler through the axioms generated by `native_decide`. The source contains no unproved placeholders or user-declared axioms.

This proves Lemma 8 on the entire admissible domain and hence completes the proof of Theorem 1. The role of the certificate is precise: it verifies the final uniform scalar inequality. The dependence on r that produces the improved bound enters earlier through (3.2) and (3.4).

AI disclosure

OpenAI Codex autonomously developed the proof, formalized it in Lean, generated the exact numerical certificates, and drafted the initial manuscript. The authors supplied the research question and candidate materials, reviewed the resulting argument, references, and verification record, and prepared this version for arXiv.

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